

SECTION C:

SEQUENCES AND SERIES

1. The first term of an arithmetic series is 3, the common difference is 4 and the sum of all term is 820. Find the number of terms and the last term.

- A. 79, 20
- B. 20, 79 +++
- C. -20.5, 80
- D. 80, -20.5

2. Find the sum to which the series converges $\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \dots$

- A. $\frac{1}{2}$
- B. $\frac{1}{4} +++$
- C. $\frac{1}{8}$
- D. $\frac{1}{16}$

3. What is the ninth term of the geometric sequence if the third term and the seventh term are -1 and -81?

- A. 3
- B. $-\frac{1}{9}$
- C. -729 +++
- D. 927

4. What is the second term of the series, if the three numbers in geometric sequence whose sum is 13 and whose product is 64

- A. 4 +++
- B. 3
- C. 5
- D. 6

5. Evaluate the tenth term of the series $3 + 9 + 27 + 81 + \dots$

- A. 50499
- B. 59490
- C. 59049 +++
- D. 59940

6. Find the three numbers in arithmetic progression whose sum is 3 and whose product is -15.

A. 5,-3,-1

B. -3,1,5 +++

C. 1,5,-3

D. None

7. If $U_1 = -1, U_2 = -5$ and $U_r = a + br$, find a and b .

A. 2,-3

B. -1,-5

C. 3,-4 +++

D. 4,-3

8. To what sum does the following series converges $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$

A. $\frac{1}{3}$

B. 3

C. $\frac{3}{4}$ +++

D. 4,-3

9. The first term and the last term of a geometric series are 3 and 768, if the sum of the terms is 1533. Find the common ratio.

A. 3

B. $\frac{1}{2}$ +++

C. 2

D. $-\frac{1}{2}$

10. Find the first term and common ratio of the geometric series if the third term and the seventh term are 81 and 16 respectively.

A. $\frac{2}{3}, \frac{729}{4}$ +++

B. $\frac{3}{2}, \frac{4}{729}$

C. $\frac{2}{3}, \frac{4}{729}$

D. $\frac{3}{2}, \frac{729}{4}$

11. The sum of the first twenty terms of an arithmetic progression is 45, and the sum of the first forty term is 290. Find the first term and the common difference.

A. $-\frac{5}{2}, 2$

B. $-\frac{2}{5}, \frac{1}{2}$

C. $\frac{1}{2}, -\frac{5}{2}$

D. $-\frac{5}{2}, \frac{1}{2}, \dots$

12. Find the formula for $12, 6, 3, \frac{3}{2}, \frac{3}{4}, \dots$

A. $n!$

B. $16 - 4n$

C. $\frac{24}{2^n}$ +++

D. $2n + 3$

13. What limit does the sequence $\left\{ \frac{n^2 + 1}{3n^2 + 4n + 2} \right\}_{n=1}^{\infty}$ converges into?

A. 0

B. ∞

C. $\frac{1}{3}$ +++

D. 1

14. Find the S_n for the arithmetic sequence $8, 12, 16, 20, \dots$

A. $4n + 4$ +++

B. $2n(n + 3)$

C. $2n(n - 1)$

D. $2n(n + 4)$

15. Find the sum of all the odd numbers between 1 and 99.

A. 25000

B. 2500 +++

C. 250

D. 250000

16. A model railway manufacturer makes pieces of track of lengths 8cm, 10cm, 12cm etc up to and including 38cm. An enthusiast buys 5 pieces of each length. What is total length of track can be made?

A. 345 +++

B. 300

C. 380

D. 400

17. The general formula for the reciprocal of geometric sequences is...

A. $\frac{r(r^n - 1)}{ar^n(r - 1)}$ +++

B. $\frac{r(r^n - 1)}{r^n(r - 1)}$

C. $\frac{(r - 1)}{ar(r^n - 1)}$

D. $\frac{a(r^n - 1)}{r^n(r - 1)}$

18. The sums $1, 1+2, 1+2+3, \dots$ referred to as

- A. Sequence
- B. triangular number +++
- C. amicable number
- D. perfect square

19. Which one of the following is correct for Harmonic progression

- A. $\frac{1}{a + (n-1)d}$ +++
- B. $a + l$
- C. ar^{n-1}
- D. $\sqrt{a + (n-1)d}$

20. Find the sum of the first twenty-five odd numbers

- A. 526
- B. 625+++
- C. 265
- D. 562

21. If the second term and fourth term of an exponential function are 6 and 54, find the n^{th} term

- A. $\frac{3^{n-1}}{2}$
- B. 3^{n-1}
- C. $2 \cdot 3^{n-1}$ +++
- D. 6^{n-1}

22. The first term of a geometric progression is 10 and the sum to infinity is 20. Find the common ratio.

- A. 2
- B. $\frac{1}{2}$ +++
- C. $\frac{1}{4}$
- D. 4

23. The first term of an A.P. is 4 while the sum to infinity is 20. Find the common ratio

- A. +++
- B. 4
- C.
- D. 5

24. If the sixth term of an arithmetic progression is 11 and the

first term is 1, find the common difference

A. $2 + + +$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. 4

25. The fourth term of an arithmetic progression is 13 while the tenth term is 31. Find the twenty-first term

A. 46

B. $64 + + +$

C. $\frac{1}{3}$

D. 3

26. The second term and the fourth term of a geometric progression are 8 and 32 respectively. What is the sum of the first four terms

A. 2

B. 4

C. $60 + + +$

D. 15

27. Express the recurring decimal $0.\overline{331}$ as a fraction in its lowest term

A. $\frac{33}{99}$

B. $\frac{331}{999} + + +$

C. $\frac{133}{999}$

D. $\frac{133}{99}$

28. For what value of x does the series $\frac{1}{1+4x} - \frac{1+x}{(1+4x)^2} + \frac{(1+x)^2}{(1+4x)^3} - \dots$

A. $\frac{1+x}{(1+4x)^2}$

B. $-\frac{1+x}{(1+4x)}$

C. $-\frac{1}{4} > x > 0, -\frac{2}{5} > x > -\frac{1}{4} + + +$

D. $-\frac{1}{4} > x > 1, -\frac{5}{2} > x > -4$

29. How many terms of the series $2 + 3 + \frac{9}{2} + \dots$ must be taken for

the sum to exceed 30?

A. 4

B. 3

C. 6 +++

D. 5

30. Insert three arithmetic mean between 3 and 19. What is the sum of the linear function.

A. 4

B. 15

C. 19

D. 55+++

31. How many terms are there in a sequence of arithmetic progression whose the sum of the first and last term are 4 and 26 is 180 respectively.

A. 12+++

B. 10

C. 4

D. 30

32. Find in the sequence . If the first three term are in arithmetic sequence and the last three are geometric sequence. Find a and b.

A. $5, \frac{25}{4}$

B. $\frac{4}{25}, 5$

C. $5, -\frac{25}{4}$

D. $\frac{25}{4}, 5$ +++

33. Find in the sequence . If the first three terms are in arithmetic sequence and the last three are in geometric sequence. Find x and b.

A. $4, \frac{25}{4}$

B. $4, \frac{15}{2}$

C. $\frac{15}{2}, 4$ +++

D. $4, -\frac{15}{2}$

34. An uncle places a sum of money in saving account for a nephew when he was born. On each succeeding birthday, the uncle deposit two times more than on the previous birthday, the total sum of the first 11 deposit is . How large was the first deposit?

- A.
- B. +++
- C.
- D.

35. A father place a sum of money in a saving account for his daughter when she was born. On each succeeding birthday, the father deposits two more than on the previous birthday without interest. The total sum of the first thirteen deposit is . How large was the last three deposit.

- A. #2591
- B. #1259
- C. #9521+++
- D. #5912

36. Given in the sequence . Find the sum of the first tenth term.

- A. 513
- B. 153
- C. 155+++
- D. 551

37. The sixth and thirteenth terms of an arithmetic progression are and respectively. Find the sum of the first twenty terms.

- A. 0
- B. 14
- C. -20+++
- D. -10

38. Find the geometric mean for consecutive term .

- A. +++
- B.
- C.
- D.

39. Find the arithmetic mean for the consecutive term .

- A.
- B. +++
- C.
- D. None

40. Find the geometric mean for consecutive term

- A. 2
- B. 4+++
- C. 8
- D. 16

41. Find the arithmetic mean for consecutive term .

- A. 2+++
- B. 7
- C. -3
- D. None

42. To what sum does the following series converges

- A.
- B. +++
- C.
- D.

43. Find the sum to which the series converges

- A.
- B. 2+++
- C. 3
- D. 4

44. Find the first term and the common difference of the three numbers in an arithmetic sequence whose sum is and whose product is .

- A.
- B. +++
- C.
- D.

45. The third term and the fourth term of an arithmetic sequence are 15 and 5. What is the first term and the common difference?

- A.
- B.
- C. +++
- D.

46. What is the first term and the common ratio of the geometric sequence whose the third and the seventh term are and ?

- A. +++
- B.
- C.
- D.

47. What is the first term and the common difference of the arithmetic sequence whose the third and the seventh term are

and ?

- A. 39,20
- B. 20,39
- C. 39,-20+++

D. -20,-39

48. Express the recurring decimal as a fraction in its lowest term

A. +++

B.

C.

D.

49. Express the recurring decimal as a fraction in its lowest term

A. +++

B.

C.

D. None

50. Find the first term of the exponential function whose common ratio of the second term and last term are and

A. +++

B.

C.

D.

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